STEEL AUTHORITY OF INDIA LIMITED

सेल SAIL

SAIL VOCATIONAL TRAINING
PROJECT REPORT

On

IT ASSET MANAGEMENT

By

SHREYAS SRIVASTAVA

B. TECH IN COMPUTER SCIENCE & ENGINEERING

Institute Of Technical Education and Research

SIKSHA O ANUSANDHAN UNIVERSITY, BHUBANESWAR

Submitted to

HRD, SAIL

Guided by

Mr. Sandeep Kumar Boral
Deputy Manager, C&IT

Steel Authority of India Limited, Bokaro Steel City

INDEX

CERTIFICATE

This is to certify that the project work entitled "IT Asset Management" has been successfully carried out by Shreyas Srivastava , URN NO: 5915525 under the guidance and supervision of Mr. Sandeep Kumar Boral as a partial fulfillment of the requirements for the completion of Vocational Training (VT) in Computer & IT for a duration of 4 weeks, from 22.06.2026 to 18.07.2026

The project was undertaken with the objective of developing an efficient and centralized web- based system for managing and monitoring organizational IT assets using Firebase and modern web technologies. During the training period, the student worked sincerely and demonstrated dedication, technical skills, and practical knowledge in analyzing, designing, and implementing the project.

We hereby certify that the work presented in this project is an original effort carried out by the student under proper guidance and supervision. The project has been submitted to the department as a part of the evaluation process for the successful completion of the vocational training program.

SIGNATURE **DATE**

ACKNOWLEDGEMENT

I am deeply grateful to all those who supported and guided me throughout the development of this project titled "IT ASSET MANAGEMENT" at the steel plant.

First and foremost, I would like to express my heartfelt thanks to the Department of Human Resources and IT, whose practical insights into asset data handling provided the foundation for identifying the real-world need for this tool. Their cooperation in sharing the asset datasets with over 2030 records across 75 departments was instrumental in the success of this project.

I extend my sincere appreciation to my project guide and mentor, Mr. Sandeep Kumar Boral, Deputy Manager, C&IT, whose guidance, technical advice, and valuable feedback greatly enhanced the functionality and usability of the system.

A special note of thanks goes to the data management and IT teams for their continuous support in testing the application with live datasets involving over 2030 asset records, which helped validate the accuracy and robustness of the system logic.

Lastly, I would like to acknowledge the open-source community and the developers of Firebase, Chart.js, Font Awesome, and the Node.js/XLSX ecosystem, without which this project would not have been possible.

This project has not only enhanced my technical skills but also provided valuable experience in solving practical challenges within an industrial data environment.

INTRODUCTION

The steel industry plays a crucial role in the economic development of any nation. It is often referred to as the backbone of modern industrial society due to its widespread applications across infrastructure, construction, transportation, manufacturing, and defense sectors.

Steel is an alloy primarily composed of iron and carbon and is valued for its high strength, durability, and versatility. The industry includes a wide range of processes — from mining and refining raw materials, to producing finished steel products in integrated steel plants or mini- mills.

Global Significance:

- The steel industry is one of the largest and most essential global industries. Countries like China, India, Japan, the U.S., and Russia are among the top steel producers.
- The demand for steel continues to grow with urbanization, industrialization, and infrastructural development.

Steel Industry in India:

- India is currently the second-largest producer of crude steel in the world. The Indian steel sector is dominated by public sector undertakings (PSUs) such as SAIL (Steel Authority of India Limited) and private players like Tata Steel and JSW Steel.
- The industry supports over 2 million jobs directly and millions more indirectly through associated industries.

Importance of Steel:

- Construction: Bridges, buildings, pipelines, railways, and roads.
- Automotive: Manufacturing of cars, trucks, and heavy vehicles.
- Defense and Aerospace: High-grade steels are used in defense equipment and aircraft.
- Energy: Used in the production and transport of oil, gas, and electricity.

Current Trends:

- Emphasis on green steel and environmental sustainability.
- Adoption of Industry 4.0 technologies like automation, data analytics, and machine learning.
- Focus on increasing energy efficiency and reducing carbon emissions.

BOKARO STEEL PLANT (BSL)

The Bokaro Steel Plant (BSL) is the fourth integrated steel plant in the public sector, established with assistance from the Soviet Union. It was initially incorporated as a limited company on 29 January 1964, and later merged with Steel Authority of India Limited (SAIL)

— first as a subsidiary and subsequently as a unit — under the Restructuring & Miscellaneous Provisions Act, 1978.

Construction officially commenced on 6 April 1968. The plant currently operates five blast furnaces with a liquid steel production capacity of 5.8 million tonnes (MT). Continuous modernization is underway, and production is expected to surpass 10 MT in the near future.

During the 1990s, BSL underwent significant industrial development, including upgrades to its refining units and installation of continuous casting machines. More recently, the plant initiated an expansion program in collaboration with POSCO (Pohang Iron and Steel Company) of South Korea.

For the establishment of the plant, approximately 64 moujas (land units, each potentially including multiple villages) were acquired. Of the total land acquired, 7,765 hectares were utilized for the steel plant infrastructure.

In terms of financial performance:

- In FY 2003–04, the plant recorded a revenue of ₹11.2 billion (approx. USD 150 million).
- By FY 2007–08, revenue had increased to ₹84.26 billion.

In the financial year 2020–21, under the leadership of Managing Director Mr. Amarendu Prakash, the Bokaro Steel Plant surpassed its production and financial targets. It reported a Profit Before Tax (PBT) of ₹2,251 crore, a remarkable rise compared to the ₹48 crore PBT during the corresponding period last year (CPLY).

OVERVIEW ROLE OF IT IN STEEL INDUSTRY

The Information Technology (IT) sector has become a transformative force in the steel industry, reshaping traditional manufacturing processes, improving efficiency, reducing costs, and enhancing decision-making. In an era of digital transformation and Industry 4.0, IT acts as a critical enabler of innovation in steel production, logistics, quality control, and workforce management.

1. Automation and Process Control

- Supervisory Control and Data Acquisition (SCADA) and Distributed Control Systems (DCS) automate and monitor critical operations like blast furnace temperatures, rolling mill speeds, and cooling systems.
- These systems ensure real-time process optimization, reduce human errors, and improve safety and reliability.

2. Enterprise Resource Planning (ERP) Systems

- Steel plants like SAIL, Tata Steel, and JSW rely on ERP platforms (e.g., SAP, Oracle) for integrated management of supply chain, inventory, procurement, maintenance, finance, and HR.
- ERP streamlines business operations by providing centralized access to real-time data and performance metrics.

3. Predictive Maintenance

- IT enables condition monitoring of machines and equipment using IoT sensors and AI algorithms.
- Predictive maintenance minimizes unplanned downtime, extends asset lifespan, and reduces operational costs.

4. Quality Assurance and Data Analytics

- Computer Vision and AI-based systems are used to inspect steel surfaces and detect flaws in real time.
- Data analytics helps in analyzing product quality, performance trends, and customer feedback to drive continuous improvement.

5. Supply Chain and Logistics Optimization

- IT tools are used for supply chain visibility, fleet tracking, inventory planning, and order fulfillment.

ABSTRACT & PROBLEM STATEMENT

Abstract:

The SAIL BSL IT Asset Management System is a web-based application developed using HTML5, CSS3, Vanilla JavaScript, Firebase Authentication, and Cloud Firestore. The system efficiently manages and tracks 2030 organizational IT assets across 75 departments within the Bokaro Steel Plant environment. The primary objective is to maintain a centralized real-time record of all IT equipment — including PCs, printers, scanners, and other devices — along with their allocation details, configuration specifications, and departmental ownership.

Designed with a dark-themed, user-friendly interface, the system simplifies asset tracking, reporting, and inventory management, reducing the dependency on manual record-keeping. The application features role-based access control (IT Admin, Dept. Admin, Staff), interactive KPI dashboards, Chart.js visualizations, quick searching, filtering across 2030 records, and paginated data tables

The project is particularly beneficial for IT administrators and management teams involved in asset monitoring, auditing, maintenance planning, and resource allocation. By digitizing the asset management process on Firebase's cloud platform, the system minimizes errors, enhances transparency, and ensures better control over organizational IT infrastructure.

Problem Statement:

In large industrial plants and organizations, managing IT assets such as PCs, printers, scanners, and other hardware devices is often performed manually using spreadsheets or physical records. This traditional approach makes it difficult to maintain accurate and up-to-date information regarding the availability, allocation, and status of assets across different departments. As the number of devices increases — as is the case with 2030 assets across 75 departments in BSL — tracking details such as system numbers, serial numbers, assigned departments, and asset ownership becomes time-consuming and error-prone.

The absence of a centralized asset management system can lead to issues such as duplicate records (146 duplicate serials detected), missing asset information (806 assets with issues), inefficient resource allocation, difficulty during audits, and delays in generating reports related to organizational IT infrastructure.

CHALLENGES

- Understanding and organizing large volumes of asset-related information was a major challenge, as different devices contain multiple attributes like system numbers, serial numbers, MAC addresses, hostnames, OS versions, RAM, and departmental details.
- Designing a well-structured Firebase Firestore schema was challenging because the system needed to maintain data consistency and store 2030 asset records in an organized and queryable manner.
- Implementing role-based access control (RBAC) across three roles — IT Admin, Dept. Admin (HR), and Staff — using Firebase Authentication required careful design of login logic, session management, and page-level view restrictions.
- Maintaining data accuracy and preventing duplicate serial number records required proper validation mechanisms. The system identified and flagged 146 duplicate serial numbers across 2030 records.
- Creating a user-friendly dark-themed interface in pure HTML5/CSS3/JavaScript without any frontend framework was challenging, requiring custom component design for sidebar navigation, pagination, modals, and search.
- Implementing efficient client-side search and multi-field filtering across 2030 records required optimized JavaScript logic to ensure fast response times.
- Generating comprehensive Chart.js 4.4 visualizations for PC model distribution, RAM distribution, OS distribution, asset health overview, and issue breakdown required careful data aggregation from Firestore.
- Bulk-creating 2030+ Firebase Authentication accounts and Firestore user profiles using Firebase Admin SDK (Node.js) with the XLSX npm package was technically complex and required error handling for duplicate accounts.
- Designing the application to be scalable and role-sensitive — where Staff see only their own asset, HR Admins see their department, and IT Admin sees everything — was architecturally demanding.

TECHNOLOGY USED

Frontend:

- HTML5 — Markup structure of all pages (dashboard, login, create-users)
- CSS3 — Styling, dark theme, responsive layout, sidebar and card components
- Vanilla JavaScript (No Framework) — All interactivity, DOM manipulation, pagination, search, and filtering
- Chart.js 4.4 — Bar charts, doughnut charts, and pie charts for data visualization
- Font Awesome 6.5 — Icon set used throughout the navigation and KPI cards

Backend / Cloud:

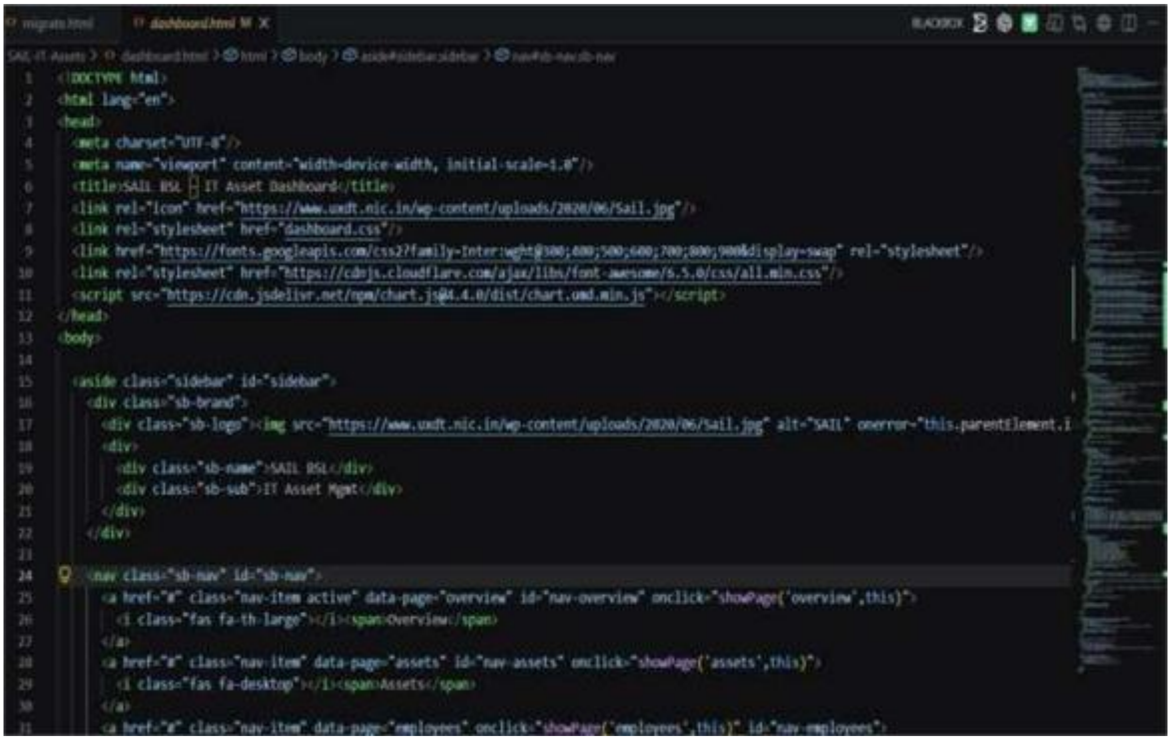
- Firebase Authentication — Login/logout system with role-based access control (IT Admin, HR Admin, Staff)
- Cloud Firestore — Real-time NoSQL cloud database storing 2030 asset records and user profiles
- Firebase Admin SDK (Node.js) — Bulk creation of Firebase Auth accounts and Firestore user profiles via admin scripts

Data & Tools:

- Node.js — Runtime for Firebase Admin SDK scripts (admin/bulk-create only, not the web app)
- XLSX (npm package) — Reading Excel files and converting asset data to JSON for Firestore upload
- Git & GitHub — Version control and project hosting

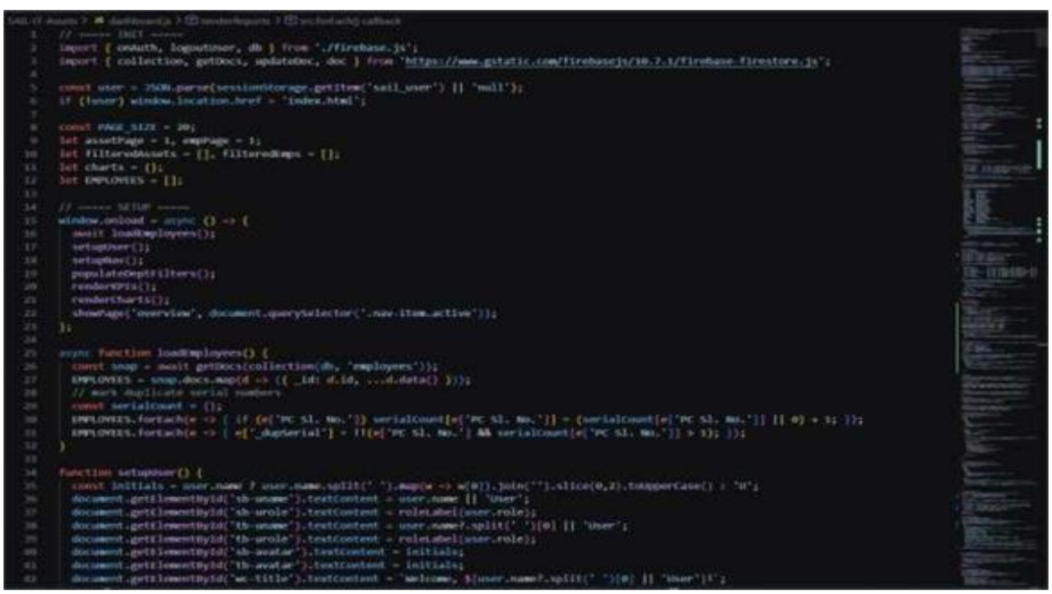
CODE SNIPPETS

1. dashboard.html — Page Structure & Navigation



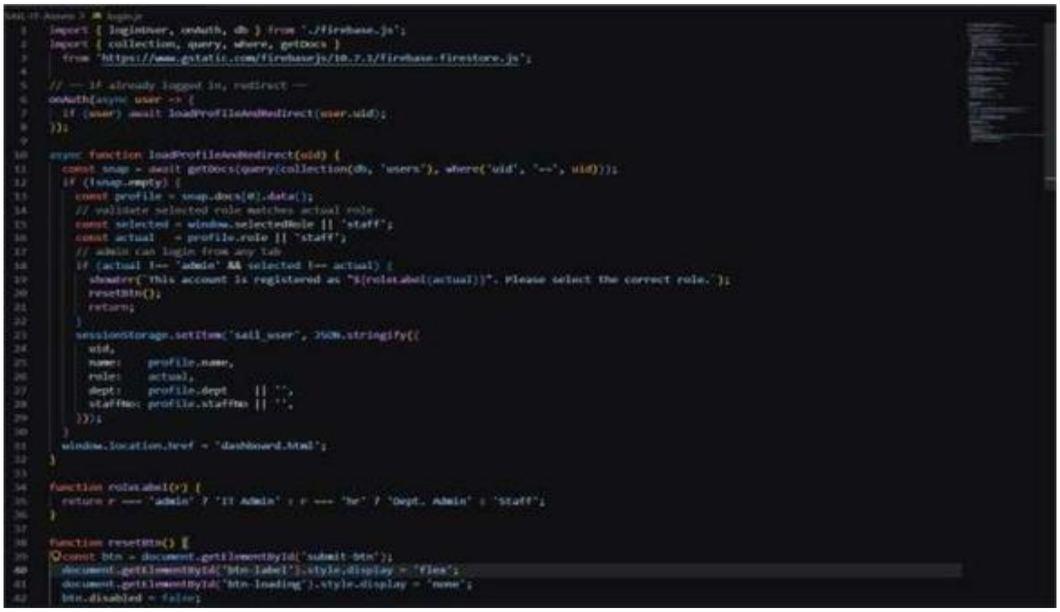
```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>SAIL BSL – IT Asset Dashboard</title>
  <link rel="icon" href="https://www.uxdt.nic.in/wp-content/uploads/2020/06/Sail.jpg"/>
  <link rel="stylesheet" href="dashboard.css"/>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800;900&display=swap" rel="stylesheet"/>
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"/>
  <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
</head>
<body>

  <aside class="sidebar" id="sidebar">
    <div class="sb-brand">
      <div class="sb-logo"><img src="https://www.uxdt.nic.in/wp-content/uploads/2020/06/Sail.jpg" alt="SAIL" onerror="this.parentElement.i
      <div>
        <div class="sb-name">SAIL BSL</div>
        <div class="sb-sub">IT Asset Mgmt</div>
      </div>
    </div>

    <nav class="sb-nav" id="sb-nav">
      <a href="#" class="nav-item active" data-page="overview" id="nav-overview" onclick="showPage('overview',this)">
        <i class="fas fa-th-large"></i><span>Overview</span>
      </a>
      <a href="#" class="nav-item" data-page="assets" id="nav-assets" onclick="showPage('assets',this)">
        <i class="fas fa-desktop"></i><span>Assets</span>
      </a>
      <a href="#" class="nav-item" data-page="employees" onclick="showPage('employees',this)" id="nav-employees">
```

2. dashboard.js — Firebase Initialization & Employee Load



```javascript
// ────── INIT ──────
import { onAuth, logoutUser, db } from './firebase.js';
import { collection, getDocs, updateDoc, doc } from 'https://www.gstatic.com/firebasejs/10.7.1/firebase-firestore.js';

const user = JSON.parse(sessionStorage.getItem('sail_user') || 'null');
if (!user) window.location.href = 'index.html';

const PAGE_SIZE = 20;
let assetPage = 1, empPage = 1;
let filteredAssets = [], filteredEmps = [];
let charts = {};
let EMPLOYEES = [];

// ────── SETUP ──────
window.onload = async () => {
  await loadEmployees();
  setupUser();
  setupNav();
  populateDeptFilters();
  renderKPIs();
  renderCharts();
  showPage('overview', document.querySelector('.nav-item.active'));
};

async function loadEmployees() {
  const snap = await getDocs(collection(db, 'employees'));
  EMPLOYEES = snap.docs.map(d => ({ _id: d.id, ...d.data() }));
  // mark duplicate serial numbers
  const serialCount = {};
  EMPLOYEES.forEach(e => { if (e['PC Sl. No.']) serialCount[e['PC Sl. No.']] = (serialCount[e['PC Sl. No.']] || 0) + 1; });
  EMPLOYEES.forEach(e => { e['_dupSerial'] = !!(e['PC Sl. No.'] && serialCount[e['PC Sl. No.']] > 1); });
}

function setupUser() {
  const initials = user.name ? user.name.split(' ').map(w => w[0]).join('').slice(0,2).toUpperCase() : 'U';
  document.getElementById('sb-uname').textContent = user.name || 'User';
  document.getElementById('sb-urole').textContent = roleLabel(user.role);
  document.getElementById('tb-uname').textContent = user.name?.split(' ')[0] || 'User';
  document.getElementById('tb-urole').textContent = roleLabel(user.role);
  document.getElementById('sb-avatar').textContent = initials;
  document.getElementById('tb-avatar').textContent = initials;
  document.getElementById('wc-title').textContent = `Welcome, ${user.name?.split(' ')[0] || 'User'}!`;
```

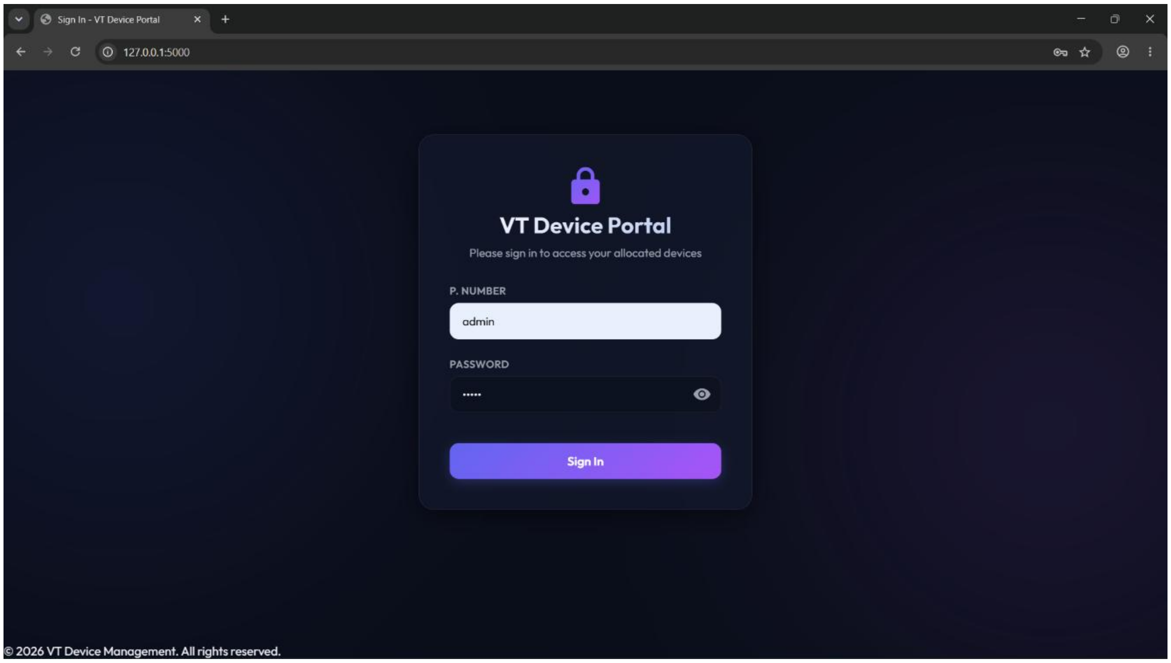
3. login.js — Role-Based Authentication



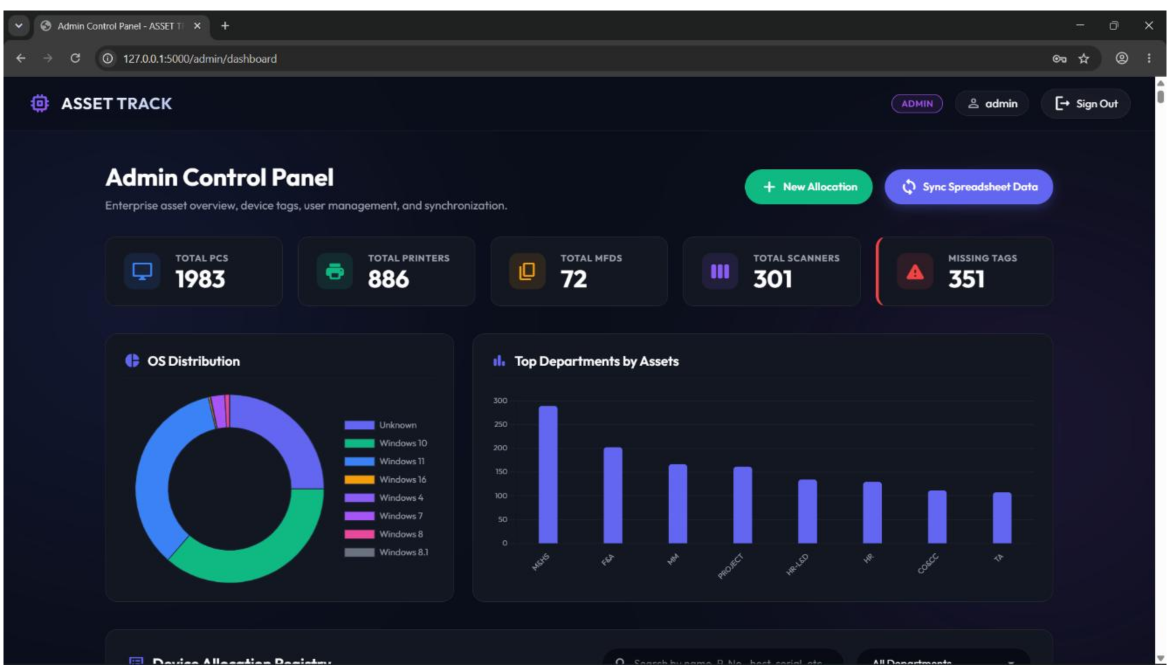
4. create-users.html — Bulk User Account Creation

OUTPUT

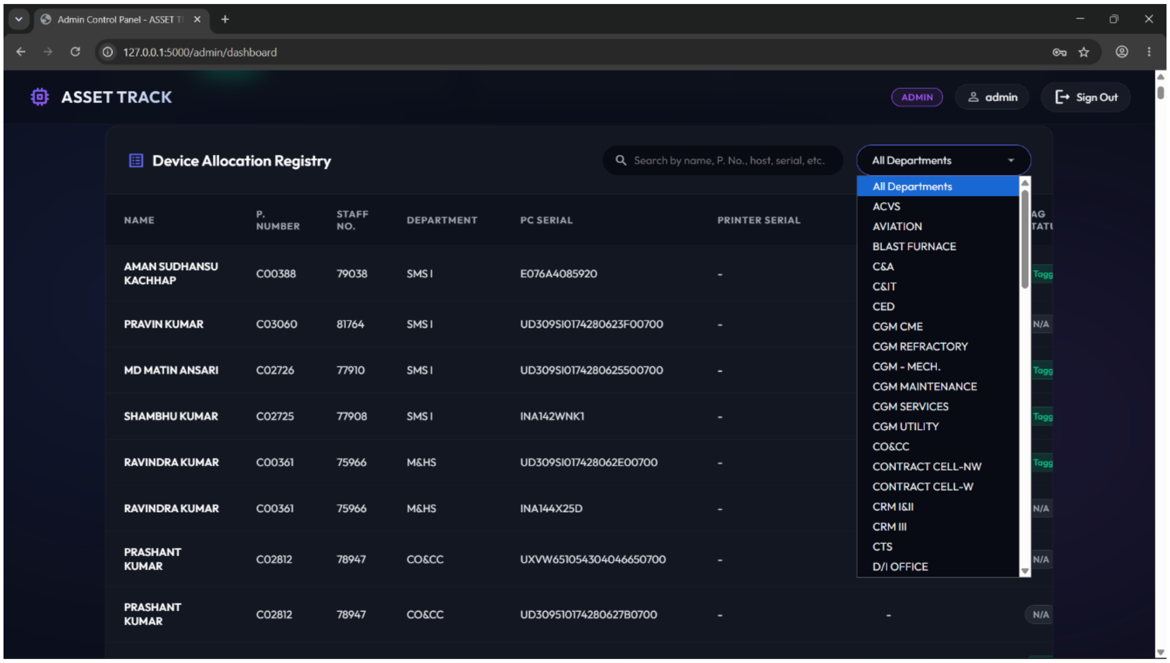
Login Page — Role-Based Secure Access Portal



Overview Dashboard — KPI Cards & Assets by Department Chart



Departments View — Per-Department Asset Cards



Reports Page — PC Model, Asset Health, RAM, OS & Printer Charts

CONCLUSION

The SAIL BSL IT Asset Management System successfully provides a centralized, real-time, and efficient cloud-based solution for managing 2030 organizational IT assets across 75 departments within the Bokaro Steel Plant. Built with HTML5, CSS3, Vanilla JavaScript, Chart.js 4.4, Font Awesome 6.5, Firebase Authentication, and Cloud Firestore, the project addresses the major limitations of traditional manual record-keeping by offering a fully digital and accessible platform.

The system's role-based access control (IT Admin, Dept. Admin, Staff) ensures that each user sees only the data relevant to their responsibilities, enhancing both security and usability. The login page features a clean role-selector UI (Staff, Dept. Admin, IT Admin) with Staff No. and password fields, built using pure HTML/CSS/JS with Firebase Auth backend. The bulk user creation utility, built using Firebase Admin SDK and Node.js with the XLSX package, enabled efficient onboarding of all employees.

Key outcomes achieved include: 100% asset tagging across all 2030 records, identification of 146 duplicate serial numbers, detection of 806 assets with data quality issues, visualization of PC model distribution (HP 8200: 625 units, ACER X4240G-AMD5: 544 units), and real-time department-wise asset breakdowns across 75 departments.

By digitizing the asset management process on Firebase's scalable cloud infrastructure, the application minimizes errors, enhances transparency, and ensures better control over organizational IT resources. The system can be further enhanced in the future with QR/barcode scanning, automated maintenance alerts, audit trail logs, and a mobile-responsive interface.

Overall, this vocational training project at SAIL BSL has been a highly valuable learning experience, providing hands-on exposure to Firebase cloud services, real-world IT asset management challenges, and full-stack web development using modern tools without any JavaScript framework.